Synaptic Plasticity as Short-Term Memory

DataForge 2026 — Pathway Track Concept Summary

Track: DataForge 2026 Pathway Track	**Topic:** Synaptic Plasticity as Short-Term Memory	**Engine:** Verified Sparse Hebbian (n=200, k=10)	**Live URL:** localhost:8000 (FastAPI)

1. Problem & Core Technical Claim

Modern artificial intelligence predominantly separates durable parameter weights (trained via gradient descent) from dynamic working memory (maintained via dedicated architectural structures like transformer KV-caches, recurrent hidden states, or explicit memory buffers). This project investigates a biologically inspired alternative: can a network with no dedicated memory component retain recent information purely through transient synaptic plasticity? Our central claim demonstrates that covariance Hebbian updates temporarily strengthen co-active synapses to form an accessible short-term memory trace. This transient state does not rely on an engineered forget gate; rather, memory fidelity degrades naturally through scalar synaptic decay, background stochastic noise, and crosstalk interference as subsequent patterns overwrite shared synaptic connections.

2. Target Learner & Pedagogical Objectives

This interactive visual learning environment is engineered for data scientists, machine learning practitioners, and students familiar with foundational neural networks (neurons, weight matrices, activations) who are new to biologically plausible memory mechanisms and plasticity-driven architectures. By engaging with the sandbox, learners achieve three concrete outcomes: (1) explaining mechanistically how covariance Hebbian plasticity transforms a weight matrix into an associative working memory; (2) predicting quantitatively and qualitatively how retention delay and distractor interference degrade pattern reconstruction; and (3) articulating the architectural bridge to frontier models such as the Dragon Hatchling (BDH), recognizing how local synaptic plasticity can serve as working memory at scale.

3. Mathematical Architecture & Real Computation

The teaching engine executes live, un-crunched linear algebra on every interaction—never relying on precomputed lookup tables or static visual animations. The network comprises n = 200 units operating in a sparse, non-negative activation regime where k = 10 units (~5% sparsity) fire per pattern, directly reflecting biological sparsity levels and reported BDH regimes. When a pattern x is presented, synapses undergo a covariance Hebbian update: delta-W = (x - p)(x - p)^T with a zeroed diagonal to prevent self-excitation, where p = 0.05 is the baseline firing probability. Over subsequent time steps, synaptic weights experience continuous passive decay W = W * (1 - lambda) coupled with symmetric Gaussian synaptic noise, while active interference occurs whenever distractor patterns are encoded into the identical synaptic substrate. Memory retrieval employs one-shot content-addressable readout: presented with a partial cue revealing 40% of the active units, the network computes raw activations via W @ cue, enforces known-mask clamping, and activates the top-k scoring units. Reconstruction accuracy is evaluated via strict active-unit intersection overlap (overlap / k), preventing sparsity artifacts from skewing performance metrics.

4. Learner Interaction & The Diagnostic Journey

The curriculum guides learners through a scaffolded 17-stage visual journey combining guided trials, predictive hypothesis testing, and diagnostic gamification. In Challenge 1, learners formulate hypotheses on distractor interference before running empirical sweeps. In Challenge 2 (Diagnose False Memory), learners analyze the reconstructed grid alongside ground truth, identifying specific hallucinated units caused by synaptic crosstalk. To guarantee complete accessibility, Challenge 2 provides both direct canvas inspection and fully keyboard-navigable unit controls (Tab, Enter, Space). In Challenge 3 (Memory Recovery), learners determine the cue completeness threshold necessary to overcome severe synaptic corruption. Challenge 4 isolates passive temporal decay from active interference, cementing the 60-second diagnostic test: explaining why memories fail without an explicit deletion operation.

5. Scientific Honesty & The Dragon Hatchling (BDH) Bridge

A critical educational requirement is absolute scientific integrity regarding what this toy model demonstrates versus what it does not. Our 200-unit network is an independent pedagogical isolation of a single concept; it is not the official Dragon Hatchling model (Kosowski et al., 2025, arXiv:2509.26507). The sandbox explicitly contrasts its dense 200x200 matrix against BDH's scale-free, heavy-tailed sparse graph; its static binary vectors against BDH's continuous spiking sequence dynamics; and its one-shot readout against BDH's formal derivation from transformer attention equations. Furthermore, we connect this mechanism to recent primary literature—including Ellwood (PLoS Comput. Biol., 2024), which demonstrates how short-term Hebbian potentiation performs computations analogous to transformer key-query attention, and Duan et al. (ICLR 2023), confirming Hebbian plasticity's efficacy for associative retention. The sandbox honors these boundaries, equipping learners to appreciate biological memory mechanisms without overclaiming equivalency to multi-parameter production architectures.

Verification & Provenance: Computational engine verified with zero precomputed shortcuts. Primary Citations: Kosowski et al. (arXiv:2509.26507, 2025); Ellwood (PLoS Comput. Biol. 20(1):e1011843, 2024); Duan et al. (ICLR 2023); Behrouz et al. (arXiv:2501.00663, 2024); Szelogowski (arXiv:2507.21474, 2025).